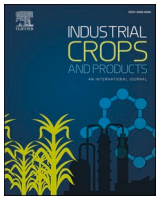
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# Deep eutectic solvent-mediated demethylation of citrus polymethoxyflavones: Effects, optimization, and bioactivity evaluation

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ABSTRACT

Deep eutectic solvent (DES)-mediated demethylation presents a novel strategy for valorizing agricultural waste. This study investigated the demethylation of polymethoxyflavones (PMFs) from citrus peel with DES and its effects on final products' biological activity. Six DESs with different functional groups ( carboxylic acid, alcohol and amide) were compared for PMF demethylation efficiency. Results showed that acidic DES effectively promoted PMF demethylation to form 5-hydroxy-polymethoxyflavones (5-OH-PMFs), with pH, hydrogen bond strength, polarity and reaction temperature being key factors. Under optimized conditions (1:5 g/L solid-liquid ratio, 480 W microwave power, 10 min), choline chloride/citric acid achieved 96.32 ± 0.06 % demethylation efficiency. Moreover, the system retained a conversion efficiency above 75 % after four consecutive cycles. Bioactivity evaluation further revealed that 5-OH-PMFs exhibited significantly enhanced antioxidant and anti-inflammatory effects compared to PMFs in RAW264.7 cells, due to increased antioxidant enzyme activity and elevated expression of anti-inflammatory factors. Simulated gastrointestinal digestion further improved 5-OH-PMF bioavailability. In summary, this study demonstrated the effectiveness of using DES to catalyze PMFs demethylation as a promising and sustainable method to enhance their bioactivity. It provides a new perspective for preparing demethylated bioactive compounds from citrus and other plant resources.

## 1. Introduction

Deep eutectic solvents (DES) are a class of novel green solvents that have garnered significant attention (Mavai et al., 2025). In recent years, DES have been widely used as novel green solvents for extracting bioactive compounds from various plant resources (Huang et al., 2024; Wu et al., 2025). Notably, DES not only efficiently extracts target compounds but also simultaneously catalyzes molecular structural transformations (Chen et al., 2023). Currently, the application of DES in the conversion of bioactive substances is primarily focused on deglycosylation reactions. Sun et al. (2020) utilized a one-pot method based on DES to extract polygonum glycosides from polygonum and simultaneously convert them into resveratrol, a process involving the cleavage of glycosidic bonds. Li et al. (2022) combined DES with pulsed electric

*Abbreviations:* PMFs, Polymethoxyflavones; 5-OH-PMFs, 5-hydroxy-polymethoxyflavones; DES, Deep eutectic solvents; HBDs, hydrogen bond donors; HBAs, hydrogen bond acceptors; ChCl, Choline chloride; OA, oxalic acid; CA, citric acid; LA, lactic acid; Gly, Glycerine; BDO, 1,4-Butanediol; UA, urea; ISO, isosinensetin; SIN, sinensetin; NOB, nobiletin; TAN, tangeretin; 5-DN, 5-Demethylnobiletin; 5-DT, 5-Demethyltangeretin; MTNE, methyl tert-butyl ether; ABTS, 2,2′-azinobis 3-ethylbenzothiazoline-6-sulfonic acid; DPPH, 1,1-diphenyl-2-picrylhydrazyl radical; TPTZ, 2,4,6-tripyridyl-s-triazine; LPS, Lipopolysaccharides; $H_2O_2$,, Hydrogen peroxide; MDA, malondialdehyde; GSH, glutathione; SOD, superoxide dismutase; CAT, catalase; GR, glutathione reductase; HPLC, high-performance liquid chromatography; NO, nitric oxide; TGF-β, transforming growth factor-beta; IL-10, interleukin-10; IL-6, interleukin-6; IL-1β, interleukin-1 beta; TNF-α, tumor necrosis factor-alpha.

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field technology to significantly enhance the conversion rate of flavonoid glycosides to flavonoid aglycones, which also falls under glycoside removal reactions. Similarly, multiple studies have confirmed the superior performance of DES in promoting the deglycosylation reactions of glycoside compounds such as soybean isoflavone glycosides (Chen et al., 2023). These studies utilize the acidic environment and hydrogen bond network structure of DES to promote the hydrolysis and cleavage of glycosidic bonds, thereby achieving the conversion of glycoside compounds into aglycones. However, the potential application of DES in other types of bioactive substance conversion reactions remains largely unexplored.

Demethylation modification plays an important role in enhancing the biological activity of natural products. Recent studies have revealed the potential of DES to promote demethylation reactions during lignocellulose processing. For instance, studies indicate that during the stepwise separation and depolymerisation of lignin by acidic DES, the demethylation of lignin structural units occurs concurrently with the cleavage of β-O-4′ ether bonds (Hong et al., 2020). However, thoses studies mainly focus on the overall conversion of biopolymeric macromolecules such as lignin, with no systematic investigation into demethylation reactions within specific bioactive compounds (Hong et al., 2020; Luo et al., 2025). It is unclear whether DESs can serve as efficient green catalysts for the selective demethylation of small-molecule bioactive substances. Inspired by the demethylation phenomena observed in lignin studies, we hypothesise that DES may offer an innovative and effective strategy for the controlled demethylation of bioactive molecules.

Polymethoxyflavones (PMFs) are characteristic secondary metabolites produced by citrus plants, exhibiting bioactivities such as antioxidant, anti-inflammatory, anticancer, and lipid metabolism regulation (Peng et al., 2022). However, the primary active forms of PMFs that exert their effects in vivo are their demethylated products. These demethylated PMFs exhibit significantly enhanced bioavailability compared to their original forms (Ho et al., 2020). This makes the demethylation of PMFs a key pathway for enhancing the functional value of citrus resources. Several studies have explored various approaches to convert PMFs to 5-OH-PMFs, including acid hydrolysis (Li et al., 2007), thermal treatment (Zhang et al., 2019a, 2019b) and microbial biotransformation (Łużny et al., 2021). Among these, Brønsted acid–based hydrolysis has emerged as the main method for producing 5-OH-PMFs (Huang et al., 2019). The reaction mechanism likely due to the following: (1) existence of lone pairs of oxygen atoms in the methoxy group of PMFs and the oxygen atom of the adjacent carbonyl group, which are prone to proton activation; (2) presence of acidic solution providing protons; (3) the proton hydrogen and electron donating oxygen coordination to form a stable six-membered ring structure; (4) the broken covalent bond between the oxygen atom and methyl group; (5) hydrolysis of the six-membered ring to form a hydroxyl group (Huang et al., 2019). However, liquid acids are difficult to recover, corrode equipment, and pollute the environment. Although heating is safer, it is inefficient and offers low product selectivity. Microbial conversion requires complex equipment and similarly exhibits poor selectivity. These limitations hinder the realization of green production goals.

Various physical techniques are also extensively used to enhance demethylation. High-temperature treatment significantly increases reaction rates. For instance, phosphoric acid-catalysed demethylation of naringin achieves 93.71 % conversion within two hours at 175 °C (Huang et al., 2019). Hot pressurised water technology enhances acid-catalysed demethylation by altering water's physicochemical properties (Bocus et al., 2025). Ultrasound-assisted technology (UAE) uses locally generated high temperatures and pressures via cavitation effects to disrupt plant cell walls and enhance mass transfer (Sun et al., 2020). Microwave-assisted technology (MAE) selectively heats polar molecules, inducing intense molecular motion and friction to accelerate chemical reactions (Zhang et al., 2025). These studies demonstrate that integrating physical enhancement techniques with DES holds considerable potential, offering diverse intensification approaches for PMF demethylation.

This study investigated the effects of DES type (carboxylic acid, alcohol and amide-based), physicochemical properties (pH, hydrogen bond strength, Kamlet-Taft parameter), and physical techniques (microwave, high temperature and pressure, ultrasound-bath) on the demethylation of PMFs. Based on this, key reaction parameters were optimized, including microwave power, reaction time, DES water content, and feed-to-liquid ratio. Additionally, the anti-inflammatory and antioxidant activities of 5-OH-PMFs were evaluated. The results of this study indicate that acidic DES have potential in the preparation of hydroxylated PMFs extracts and other demethylated bioactive compounds.

## 2. Methods

### 2.1. Chemicals

Choline chloride (ChCl, ≥ 98 %), oxalic acid (OA, ≥ 98 %), citric acid (CA, ≥ 99 %), lactic acid (LA, ≥ 99 %), Glycerine (Gly, ≥ 98 %), 1,4-Butanediol (BDO, ≥ 99 %), urea (UA, ≥ 99 %) and acetic acid (≥ 98 %) were purchased from Macklin Biochemical Technology Co., Ltd (Shanghai, China). Standards included isosinensetin (ISO), sinensetin (SIN), nobiletin (NOB), tangeretin (TAN), 5-Demethylnobiletin (5-DN) and 5-Demethyltangeretin (5-DT), all sourced from Yuanye Bio-Technology Co., Ltd (Shanghai, China). Sigma Aldrich (St. Louis, MO, USA) provided Folin-Ciocalteu reagent, acetonitrile, Petroleum ether, methyl tert-butyl ether (MTBE), 2,2′-azinobis 3-ethylbenzothiazoline-6-sulfonic acid (ABTS), 1,1-diphenyl-2-picrylhydrazyl radical (DPPH), 2,4,6-tripyridyl-s-triazine (TPTZ). Lipopolysaccharides (LPS) were purchased from Nanjing Herbal Bio-technology Co., Ltd (Nanjing, China). Hydrogen peroxide ($H_2O_2$) was purchased from Chongqing Chuandong Chemical Co., Ltd (Chongqing, China). Pepsin, Trypsin, Pancreatic lipase and bile salts were purchased from Macklin Biochemical Technology Co., Ltd (Shanghai, China).

### 2.2. Materials

Mature citrus fruits (*Citrus reticulata* Blanco cv.'Nanfeng tangerine') were collected from the National Citrus Germplasm Repository (Chongqing, China). Healthy fruits of average size at maturity were harvested from 5 trees per variety. The peel was peeled and rinsed with water and then air-dried at 50 ℃. The dried samples were ground into powder and filtered through a 60-mesh sieve. The powder obtained was stored in a dry and light-proof environment until use.

RAW264.7 mouse macrophages and specific culture medium were purchased from Wuhan ProCells Biomedical Technology Co., Ltd. (Wuhan, China). Malondialdehyde kit, glutathione kit, superoxide dismutase kit, catalase kit, and glutathione reductase kit were purchased from Beijing Solarbio Science & Technology Co., Ltd. (Beijing, China). RNA extraction kit, RNA reverse transcription, and quantitative kits were purchased from Beijing TIANGEN Biochemical Technology Co., Ltd. (Beijing, China).

### 2.3. PMF extraction and analysis

PMFs were extracted from citrus peels following the method previously described by Zheng et al. (2022). In brief, petroleum ether is used to extract crude PMFs, which are then washed with MTBE to obtain PMFs extracts. The extracts are stored in dry, dark bottles for subsequent use. The identification and quantification of PMFs were conducted according to the method described by Chen et al. (2022). Briefly, chromatographic analysis was performed on a Waters HPLC instrument equipped with a photodiode array (PDA) detector (Milford, MA, USA) and a SunFire C18 chromatographic column (4.6 × 250 mm, 5μm, USA). The column temperature was set at 40 °C. 0.01 % acetic acid and acetonitrile were used as mobile phases A and B. The flow rate was set at

1 mL/min. The gradient elution process was as follows: from 0 to 5 min, the proportion of B increased from 10 % to 40 %; from 5 to 10 min, the proportion of B increased from 40 % to 60 %; from 10 to 15 min, the proportion of B increased from 60 % to 80 %; from 15 to 20 min, the proportion of B increased from 80 % to 100 %; from 20 to 23 min, the proportion of B decreased from 100 % to 40 %; from 23 to 25 min, the proportion of B decreased from 40 % to 10 %; from 25 to 26 min, the proportion of B remained at 10 %. Absorbance was measured at 330 nm, and the standard curve of PMF monomer is shown in supplementary Table S1. The PMF monomers were characterized using the UPLC-QqQ-MS/MS technique, which follows the method of Zheng et al. (2022) with appropriate modifications. The final converted products were detected using a non-targeted metabolomics approach, with the procedure strictly adhering to our previous research methodology (Ye et al., 2025).

### 2.4. DES preparation and characterization

ChCl was utilized as a general HBA, while carboxylic acids—OA, CA, and LA—alcohols such as Gly and BDO, and the UA were employed as HBDs. These were mixed with the HBA at various molar ratios. The preparation of all DES was carried out by heating, following the method outlined by Abbott et al. (2004). Relevant details regarding the compositions are summarized in supplementary Table S2.

The chemical functional groups of DES were characterized using Fourier Transform Infrared Spectroscopy (FT-IR) (Shimadzu IRTracer-100, Japan) in ATR mode, covering a scanning range from 400 to 4000 $cm^{-1}$ . Thermogravimetric analysis (TG) was performed using an SDT 650 thermoanalyser (TA Instruments). Thermograms were recorded in a nitrogen atmosphere with a heating rate of 10 °C/min from 30 to 600 °C. $^1H$ NMR spectra were recorded on a JNM-ECZ600R spectrometer. Samples were dissolved in deuterium oxide ($D_2O$) prior to analysis. Each spectrum was acquired with 1024 scans and a relaxation delay of 2 s. The viscosity of DES was measured using a MCR92 rotational rheometer (Anton Paar, Austria). The lower plate temperature was regulated by a conical plate temperature control system and steady shear viscosity measurements were performed at 25 °C and 150 °C with a shear rate of 50 $s^{-1}$ . The pH of DES was determined using a pH meter (PHSJ-5, INESA, China). For solvent color development parameters (π*, β and α) of DES, N,N-diethyl-4-nitroaniline, 4-nitroaniline, and Nile red were used as dye indicators for determination, respectively. The parameters for solvent color development were calculated according to Zhang et al. (2023).

### 2.5. PMF Processing with DES

To evaluate the conversion rate and feasibility of DES in the transformation of PMFs into 5-OH-PMFs, this study selected different types of DES (carboxylic acid, alcohol and amide-based) and assessed their conversion performance under various conditions, including ultrasonic bath, microwave, and high-temperature and pressure treatments. Subsequently, Comprehensive optimization experiment was conducted to investigate the main parameters influencing the conversion process. The parameters examined included power levels of 160, 320, 480, 640 and 800 W; material-to-liquid ratios (g/L) of 1:5, 1:2, 1:1, 2:1, and 5:1; reaction times of 2, 4, 6, 8 and 10 min; and DES water contents (%) of 0, 5, 10, 15, and 20.

Following the conversion reaction, the DES solution was mixed with an equal volume of ultrapure water and stored at 4 °C for 24 h. The solution was then centrifuged to collect the precipitate. The analysis of 5-OH-PMFs was performed using the method described in 2.3. The conversion efficiency (CE) of PMFs into 5-OH-PMFs was calculated using the following formula:

$$Conversion\ efficiency(CE\%) = \frac{C_{ahp} - C_{ohp}}{C_{op}} \times \frac{M_{wp}}{M_{whp}} \times 100\%$$

where $C_{op}$ and $C_{ohp}$ represent the original contents of PMFs and 5-OH-PMFs in the extract (mg/g), $C_{ahp}$ represents the content of 5-OH-PMFs in the product after the reaction (mg/g), $M_{wp}$ and and $M_{whp}$ denote the molecular weights of the respective substances (g/mol). The CE was ultimately calculated based on the actual substances participating in the conversion process.

### 2.6. Recovery of DESs

The DES recycling process was informed by the methodology described in the previous study (Wen et al., 2024). Briefly, an equal volume of water was added to the used DES solution, and the mixture was centrifuged to collect the supernatant. The collected supernatant was then transferred to a rotary evaporator to remove water, enabling the recovery of the DES solution for reuse.

### 2.7. Biological activity assessment

#### 2.7.1. Chemical Antioxidant Activity

The chemical antioxidant capacity of the samples was evaluated using ABTS, DPPH, and FRAP assays, following the method described by Wang et al. (2025). The results were expressed as μmol Trolox equivalent (TE) per gram of extract, a standard unit for assessing antioxidant activity.

#### 2.7.2. Cellular antioxidant activity

The cytotoxicity of PMFs and 5-OH-PMFs was evaluated using the CCK-8 assay in accordance with the manufacturer's instructions to screen for the optimal dosage. An oxidative stress model was then established by treating cultured cells with 0.03 % $H_2O_2$. PMFs and 5-OH-PMFs were first dissolved in DMSO and subsequently diluted to a fixed concentration using complete culture medium. After 20 h of treatment, cell samples were collected for further biochemical analysis. The antioxidant capacity of the extracts was assessed using oxidative stress markers, including malondialdehyde (MDA), glutathione (GSH), superoxide dismutase (SOD), catalase (CAT), and glutathione reductase (GR). All assays were performed in strict accordance with the manufacturer's protocols.

#### 2.7.3. Cell anti-inflammatory activity

RAW264.7 cells were seeded into 6-well plates at an initial density of $3 \times 10^5$ cells per well and cultured for 24 h at 37 °C in a 5 % CO2 incubator. The cells were then pretreated with PMFs, 5-OH-PMFs, or DMSO for 24 h. Subsequently, fresh culture medium containing 3 μg/mL LPS or without LPS was added, and the cells were incubated for an additional 4 h. Following incubation, the cells were collected to measure nitric oxide (NO) levels and to analyze the mRNA expression levels of pro-inflammatory cytokines (IL-6, TNF-α, IL-1β) and anti-inflammatory cytokines (IL-10, TGF-β). RNA extraction and reverse transcription were performed according to the manufacturer's instructions. Quantitative analysis was carried out using the SYBR Green ProTaq HS Premix qPCR kit (Vazyme, Nanjing, China). Gene-specific primers were synthesized by UW Genetics (Shenzhen, China) and are listed in Supplementary Table S3.

#### 2.7.4. In vitro digestion

The in vitro digestion process was conducted following the protocol outlined by Brodkorb et al. (2019). The digestion buffers for the simulated oral phase (1.25 ×SSF), gastric phase (1.25 ×SGF), and intestinal phase (1.25 ×SIF) consisted of KCl, $KH_2PO_4$, $NaHCO_3$, NaCl, $MgCl_2$, $(NH_4)CO_3$, $CaCl_2$, HCl, and $ddH_2O$. Simulated saliva (SSF) was prepared using 1.25 ×SSF and $ddH_2O$. Simulated gastric fluid (SGF) included 1.25 ×SGF, pepsin, HCl, and $ddH_2O$, while simulated intestinal fluid (SIF) comprised 1.25 ×SIF, Trypsin, Pancreatic lipase, bile salts, NaOH, and $ddH_2O$. 1 M HCl and 1 M NaOH are used to adjust the pH of the

digestion solution, with the volumes recorded.

Prior to digestion, 40 mg of 5-OH-PMFs and PMFs were suspended in 5 mL of PBS. During the oral digestion phase, 5 mL of the suspension was mixed with 5 mL of SSF at a 1:1 (v/v) ratio and incubated at pH 7.0, 37 °C, and 100 rpm for 2 min. Subsequently, an equal volume of SGF was added to the mixture, followed by incubation at pH 3.0, 37 °C, and 100 rpm for 2 h to simulate gastric digestion. Finally, an equal volume of SIF was introduced to the digestion solution, and the mixture was incubated at pH 7.0, 37 °C, and 100 rpm for an additional 2 h to simulate intestinal digestion. After each digestion phase, 2 mL of the digestion solution containing terminal samples was collected, centrifuged at 3000 ×g for 10 min to obtain the supernatant, and filtered through a 0.22 μm membrane. The filtered supernatants were then analyzed using HPLC to quantify the content of 5-OH-PMFs (5-DN/5-DT) and PMFs (NOB/TAN). Solubility for each digestion phase was calculated based on the HPLC results.

### 2.8. Data statistical analysis

Data are expressed as mean ± standard deviation and repeated at least 3 times. One-way analysis of variance (ANOVA) and Duncan's test were performed using SPSS 20.0 software, with a significance level set at $p < 0.05$ to assess statistical significance. R was used for correlation analysis.

## 3. Results and Discussion

### 3.1. Preparation and characterization of citrus PMFs and synthetic DES

#### 3.1.1. Enrichment and analysis of PMFs from citrus

A high-purity mixture of PMFs from citrus peels was prepared according to the strategy illustrated in Fig. 1A. A two-step method was utilized for the selective isolation of the target PMFs compounds. The crude extract was first obtained by reflux extraction using petroleum ether. This solvent was chosen due to prior research that identified it as the most effective for extracting PMFs from citrus peel (Zheng et al., 2022). However, as crude PMFs extract typically contain lipid-soluble impurities such as essential oils and carotenoids (Mahato et al., 2019). Consequently, subsequent purification by washing with MTBE effectively removed residual impurities (Matyash et al., 2008).

The purified PMFs extract was then subjected to qualitative and quantitative analysis using UPLC-QqQ-MS/MS and HPLC-DAD,

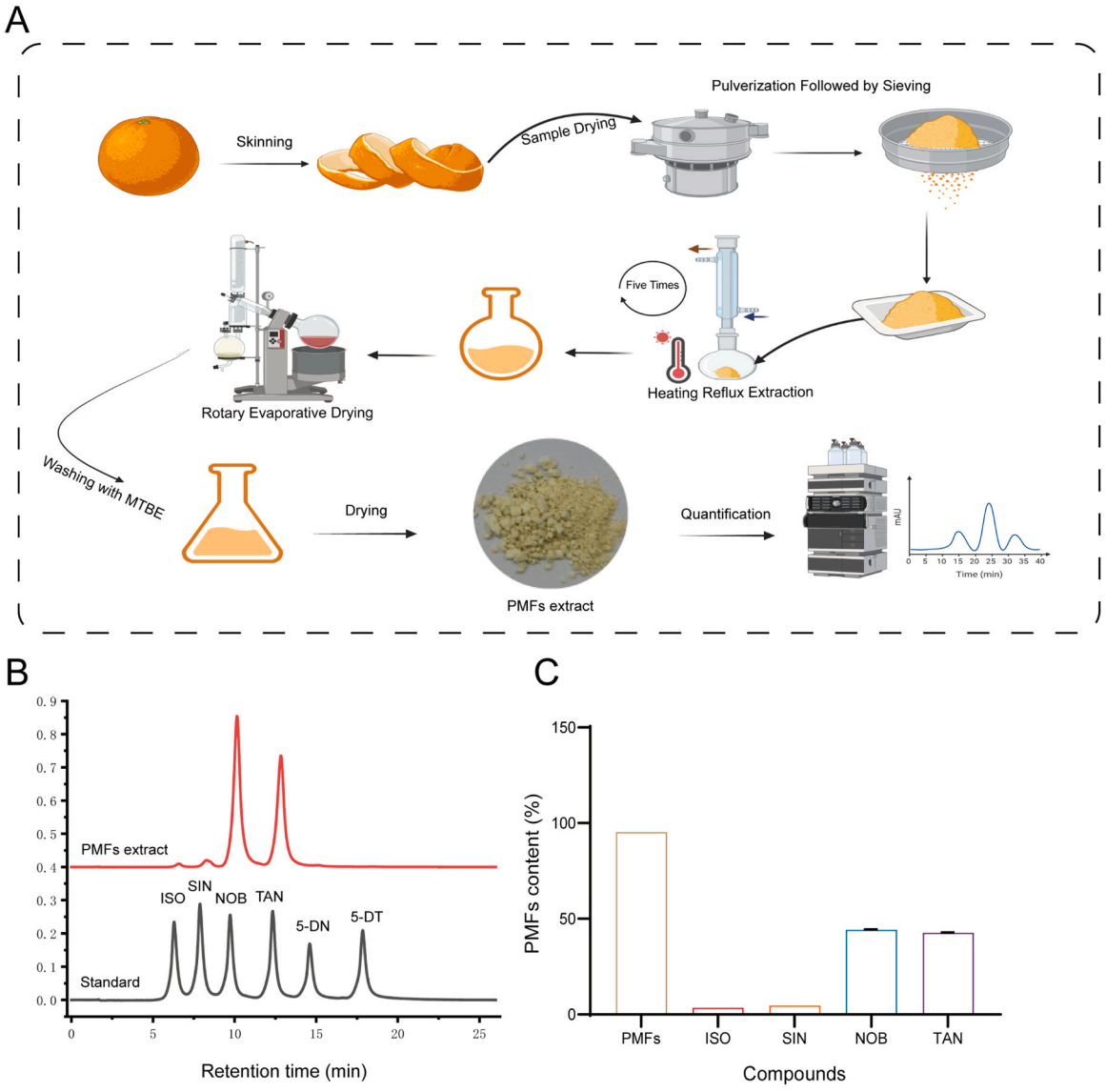



**Fig. 1.** Schematic representation of PMFs extraction and analysis from citrus peels. A Diagram of the PMFs enrichment process. B Chromatogram of enriched PMFs extract. C Content of PMFs extract.

respectively. The UPLC-QqQ-MS/MS results were presented in supplementary Table S4. It showed that four PMFs were separated and identified in citrus peel PMFs extract, including Isosinensetin (ISO), Sinensetin (SIN), Nobiletin (NOB) and Tangeretin (TAN). Then the total PMFs content was quantified using HPLC-DAD in accordance with the standard curve of the reference standard. The quantitative analysis indicated that the total PMFs content in the citrus peel PMFs extract was 95.29 ± 0.01 %. Nobiletin (44.34 %) and tangeretin (42.74 %) were the most abundant components, followed by sinensetin (4.69 %) and isosinensetin (3.52 %) (Fig. 1B and C).

### 3.1.2. Synthesis and Characterization of DES

Fourier transform infrared (FT-IR) spectroscopy is extensively utilised in the examination of molecular interactions, as it facilitates the characterisation of samples through the analysis of atomic bonds or intramolecular functional groups. Accordingly, infrared detection was carried out on these six DES solutions. As seen in Fig. 2. The O-H stretching band of ChCl was identified at 3217 $cm^{-1}$, while the $CH_2$ bending vibrational peak was observed at 1480 $cm^{-1}$ (Fig. 2B). Additionally, the characteristic vibrational peaks of C-N were detected at 1376 and 892 $cm^{-1}$, consistent with previous research (Shafie et al., 2019). The characteristics of pure fractions as HBDs are evident in their spectra: lactic acid exhibits a C═O stretching band of the carboxylate

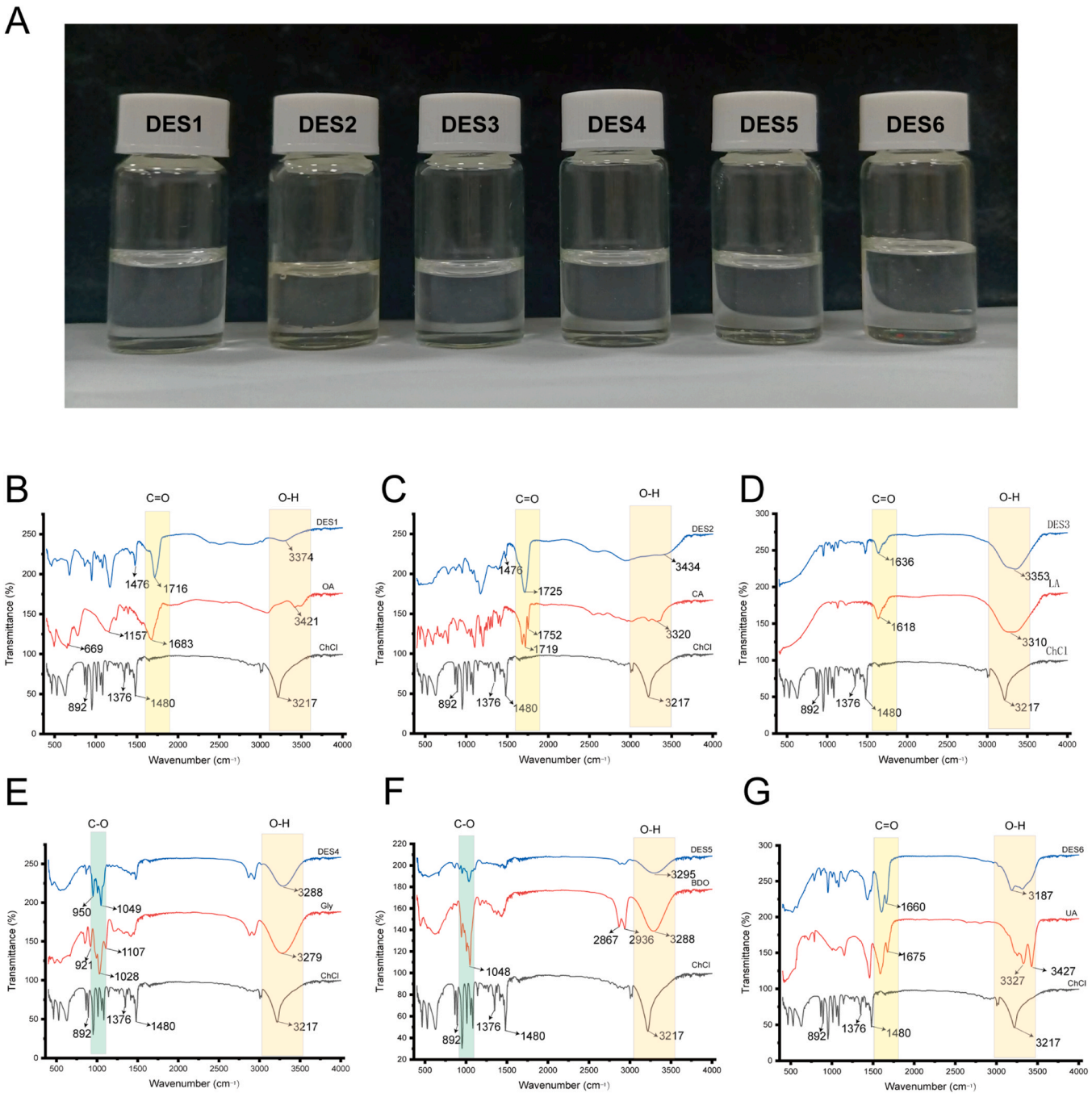



**Fig. 2.** FTIR spectra of six DES solutions. A Morphology of the six synthesised DES at room temperature. B FTIR spectra for pure ChCl, pure OA, and ChCl/OA. C FTIR spectra for pure ChCl, pure CA, and ChCl/CA. D FTIR spectra for pure ChCl, pure LA, and ChCl/LA. E FTIR spectra for pure ChCl, pure Gly, and ChCl/Gly. F FTIR spectra for pure ChCl, pure BDO, and ChCl/BDO. G FTIR spectra for pure ChCl, pure UA, and ChCl/UA.

group at 1618 $cm^{-1}$ (Fig. 2D); glycerol shows an extremely broad O-H stretching band at 3279 $cm^{-1}$ (Fig. 2E); and urea displays sharp N-H stretching vibrations at 3427 and 3327 $cm^{-1}$, along with a C═O (amide I band) absorption peak at approximately 1675 $cm^{-1}$ (Fig. 2G).

Spectroscopy revealed that all synthesised DES displayed a notably broad absorption peak (3200–3500 $cm^{-1}$), indicative of O-H bonding, characteristic of hydrogen bond formation, corroborating findings in the previous study (Ozturk et al., 2018). This broad peak also overlaps with the N-H stretching vibrational peak, which is usually found about 3000–3400 $cm^{-1}$ for systems containing urea (AlOmar et al., 2016). The C═O stretching vibrational peak of all carboxylic acid-based DES exhibited a significant shift (e.g., from 1683 to 1716 $cm^{-1}$ in the oxalic acid system), indicating a disruption of the acid dimer structure (Fig. 2B-D). The peak positions and shapes of the O-H bands in alcohol-based DES were modified, further confirming the reorganisation of the hydrogen bonding network (Fig. 2E-F) (Jiang et al., 2020). The most significant alteration was observed in the ChCl/Urea system, where the N-H peak of urea exhibited a notable redshift and broadening, resulting in its merging with the O-H peak of ChCl to create a single broad peak (Fig. 2G). Additionally, the C═O peak also experienced a redshift. The significant redshift of the N-H peak indicates that the $Cl^-$ ion in ChCl facilitates a strong hydrogen bonding interaction with the N-H group in urea, leading to the establishment of an N-H–$Cl^-$ bond, which is crucial for the formation of deep eutectic solvents (Ashworth et al., 2016). The spectral alterations in these essential functional groups offer definitive proof of the successful synthesis of all six DES.

### 3.1.3. Thermogravimetric analysis

TGA is one of the most commonly used techniques to analyse thermal properties of deep eutectic solvents. This efficient method allowed us to describe the sample's weight loss in dependence of the temperature used during the measurements. Supplementary Figure S2 reveals that there has been a progressive increase in temperature of 5 % weight loss (T5 %), temperature of 10 % weight loss (T10 %), and temperature of 50 % weight loss (T50 %) values across different DES formulations (Supplementary Figure S2A). It is important to mention that its values strongly depend on the water content of the obtained DES samples, with most DES formulations showing only 5 % weight loss at 100 °C. However, DES3 composed of ChCl/LA exhibited rapid weight loss, which may be attributed to its higher water content where the swift evaporation of water causes significant weight loss at early temperatures. Next stages are designated to the thermal decomposition of DES components. In nearly all DES samples, T50 % values exceeded 200 °C, including DES3 (Figure S2A). TGA results of weight loss at 150 °C for all DES samples can be observed in Supplementary Figure S2B. Based on these results it can be noted that all DES samples retained at least 90 % of their weight, with the exception of the DES3 sample. Therefore, the obtained results of the TGA analysis indicate high thermal stability in the analysed temperature range.

### 3.1.4. Physicochemical Properties of the Synthesized DES

Table 1 illustrates the significant variation in pH values among six DES. Notably, DES1 and DES2 exhibit strong acidity, DES3 is weakly acidic, DES4 and DES5 are nearly neutral, while DES6 demonstrates alkalinity. These pH differences are mainly due to the chemical properties of the HBDs. Organic acids (OA, CA, LA), characterized by their carboxyl groups with proton-donating capabilities, impart acidity to the DES. In contrast, urea, with its amino groups capable of accepting protons, contributes to an alkaline pH.

**Table 1**
Summary of physicochemical characterization of the six DES.

| No | DES | pH | Viscosity (mPa·s) | | Kamlet-Taft parameters | | |
|---|---|---|---|---|---|---|---|
| | | | 25 ℃ | 150 ℃ | π* | α | β |
| DES1 | ChCl/OA | 0.13 | 6389.7 | 52.2 | 0.89 | 2.53 | 0.57 |
| DES2 | ChCl/CA | 0.28 | 5788.7 | 23.0 | 1.06 | 2.24 | 0.54 |
| DES3 | ChCl/LA | 1.38 | 165 | 4.48 | 0.93 | 1.32 | 0.45 |
| DES4 | ChCl/Gly | 4.45 | 277.4 | 9.74 | 1.59 | 0.95 | 0.65 |
| DES5 | ChCl/BDO | 5.74 | 100.03 | 5.55 | 1.34 | 0.74 | 0.74 |
| DES6 | ChCl/UA | 8.43 | 903.67 | 9.10 | 1.50 | 0.85 | 0.32 |

The viscosity analysis showed that DES1 exhibited the highest viscosity (6389.7 mPa·s), followed by DES2 (5788.7 mPa·s), while DES5 had the lowest viscosity (100.03 mPa·s) at 25 °C. In general, the high viscosity of DES solutions is due to the formation of a dense network of hydrogen bonds between the components, which restricts the mobility of the free material (Abbott, 2004). From Table 1, it could be observed that carboxylic acid-based deep eutectic solvents (DES1-DES3) demonstrate higher viscosity at room temperature, indicating strong hydrogen bond interactions between ChCl and carboxylic acid. At a temperature of 150 °C, the viscosity of all DES significantly decreases, with DES1 and DES2 exhibiting the most pronounced reductions, achieving viscosities of 52.2 mPa·s and 23.0 mPa·s, respectively. This observation is consistent with the Arrhenius equation, which suggests that increased temperatures diminish intermolecular hydrogen bond interactions, consequently improving molecular mobility.

As shown in Table 1, carboxylic acid-based DES (DES1-DES3) exhibit higher α values than alcohol-based and amide-based DES, reflecting stronger hydrogen bond acidity due to the presence of organic acids (Zhang et al., 2023). DES1 shows the highest α value (2.53), correlating with its lowest pH, indicating oxalic acid's stronger acidity compared to citric and lactic acids. ChCl/alcohol DES had higher β-values than ChCl/carboxylic acid DES, which is consistent with previous studies (Zhang et al., 2023). In general, the β-value of DES is correlated with the length of the alkyl chain of the polyol (Zhang et al., 2023). Among them, DES5 has the highest β value, which may be due to the fact that 1,4-butanediol has the longest alkyl chain.The π* parameter highlights that higher values indicate stronger polarity, with ChCl-acid DES exhibiting π* values around 1.10, in line with previous findings (Zhang et al., 2023). The ChCl/alcohol DES (DES4-DES5) exhibit elevated π* values attributable to the presence of polar hydroxyl groups in glycerol and 1, 4-butanediol, resulting in high molecular polarity.

## 3.2. Effect of DES and processing methods on demethylation of PMFs

### 3.2.1. Screening of DES

The HPLC profiles of PMFs treated with different DES under microwave conditions are presented in Fig. 3A. In DES1 and DES2 systems, the peaks for NOB and TAN were noticeably diminished, while newly formed peaks, identified as 5-DN and 5-DT, were confirmed through HPLC standard comparison. Monomer content analysis revealed a decrease in NOB and TAN concentrations, accompanied by an increase in 5-DN and 5-DT levels (Supplementary Figure S1). UPLC-QqQ-MS/MS analysis further validated their production using ionic fragments from standards (Supplementary Table S5&S6). Interestingly, no 5-DN or 5-DT was detected in alcohol- or amide-based DES systems, suggesting that PMF demethylation relies on acid hydrolysis. It is known that acid-based catalysts can facilitate demethylation reactions, such as the $FeCl_3$-mediated tartaric acid-DES system for lignin conversion, which removes methoxy groups and forms phenolic hydroxyl groups (Luo et al., 2025). The demethylation mechanism, often involving the co-O-activated $SN_2$ pathway, begins with the protonation of the methoxy oxygen atom followed by nucleophilic attack (Huang et al., 2019; Bocus et al., 2025). Carboxylic acid-based hydrogen bond donors in acidic DES provide reactive protons that catalyze ether bond cleavage, and Lewis acids enhance reaction efficiency by increasing hydrogen bond acceptor strength and acidity (Luo et al., 2025). This study found that DES containing oxalic and citric acids could catalyze demethylation, while DES with lactic acid could not, due to differences in proton strength. DES with higher Kamlet-Taft α values, such as those with oxalic and citric acids, exhibit stronger proton-providing capabilities, driving demethylation reactions. In contrast, alcohol- and amide-based DES, with weaker

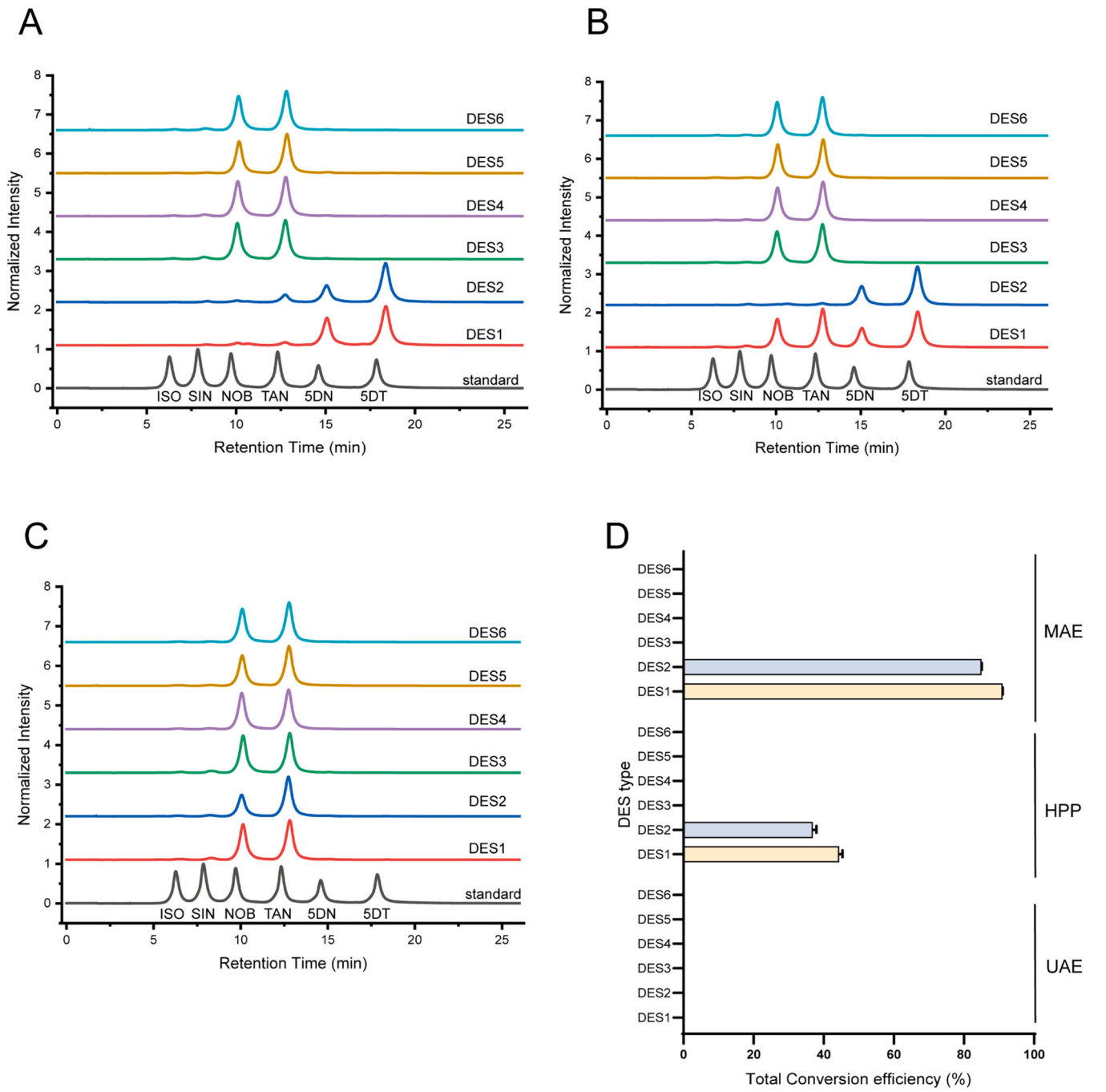



**Fig. 3.** Comparison of DES performance in hydroxylating PMFs under different conditions. A–C HPLC chromatograms of PMFs under microwave conditions (A), high-temperature and pressure (B), and ultrasound-bath conditions (C). D Conversion efficiencies of PMFs.

proton-providing abilities, were unsuitable for demethylation but more appropriate for extraction or analytical applications that preserve PMF composition. The pKa values of the HBDs further support this reasoning; the first pKa of oxalic acid and citric acid are approximately 1.25 and 2.63, respectively, which are considerably lower than that of lactic acid (approx. 3.86). Furthermore, the pKa values for alcohol-based HBDs like glycerol and 1,4-butanediol are around 14.40, indicating much weaker acidity. These findings suggest that strong carboxylic acid-based DES are ideal for catalyzing the formation of 5-OH-PMFs, while alcohol- and amide-based DES are better suited for non-conversion applications.

### 3.2.2. *Screening of processing methods*

The effects of different conditions—microwave, high-temperature and pressure and ultrasonic bath—on the demethylation of PMFs in DES systems were evaluated, as shown in Fig. 3. HPLC spectra revealed characteristic peaks of 5-DN and 5-DT in systems treated with microwave and high-temperature and pressure (Fig. 3A and B), while ultrasonic bath-treated systems showed no significant changes in PMF peaks (Fig. 3C). Concentration analysis during the transformation process confirmed that the total amount of substances remained stable, and UPLC-QqQ-MS/MS results further validated the conversion of PMFs into their corresponding 5-OH-PMFs (Supplementary Figure S1 and Tables S5-S8). Previous studies have established that acidic DES systems are involved in temperature-dependent bond-breaking reactions. For instance, hydrolysis of genistein catalyzed by ChCl/CA requires temperatures above 60 °C, with conversion efficiency positively correlated

to temperature (Chen et al., 2023). Similarly, ChCl/$TfOH_2$-mediated conversion of rosin to polymerized rosin necessitates a reaction temperature of 110 °C (Zheng et al., 2024). Consistent with these findings, microwave and high-temperature and pressure approaches rapidly increase the DES system's temperature to facilitate the reaction, unlike ultrasonic bath, which fails to achieve the required thermal conditions. Generally, ultrasonic-based methods can also provide some heat. For example, when using ultrasonic probe, since the ultrasonic probe is in direct contact with the solvent, it can better induce the cavitation effect of the solvent and rapidly heat the solvent system (Huang et al., 2025). However, in this study, we employed the ultrasonic bath method. This indirect contact resulted in an insufficient temperature rise in the DES system. It is theoretically known that an ultrasonic probe, which provides direct sonication, can generate more intense cavitation and a significant localized heating effect. This distinction may explain the limited thermal effects observed in our experimental setup, which utilized only the ultrasonic bath. For example, O-demethylation of guaiacol using Brønsted acid in high-temperature and pressure systems and phenolics production from lignin in corn kernels using DES both require high-energy input (Bocus et al., 2025; Lu et al., 2025). These results underscore the necessity of elevated temperatures for initiating PMF demethylation reactions in strongly carboxylic acid-based DES systems. It is important to note that the use of acidic DES for PMF extraction may alter PMF composition, particularly at higher temperatures, and this should be carefully considered during application.

### 3.2.3. *Effect of physicochemical properties of DES on PMF conversion efficiency*

To explore the impact of DES physicochemical properties on the demethylation conversion efficiency (CE) of PMFs, pearson correlation coefficients between various parameters and CE were analyzed (Fig. 4). A very strong negative correlation was observed between pH and CE ($r = -0.74$, $p = 5e\text{-}04$), indicating that acidic conditions significantly enhance the demethylation reaction. This is due to low pH environments favoring protonation reactions, which activate methyl-oxygen bonds and accelerate demethylation. In addition, previous studies have indicated that the ChCl/CA system can form more hydrogen bonds with solutes, with an increased hydrogen bond lifetime compared to other solvents (Huang et al., 2024). This suggests that acidic DES systems may have a higher propensity to interact with PMFs, thereby enhancing the likelihood of demethylation reactions. Furthermore, DES viscosity demonstrated a perfect positive correlation with CE ($r = 0.87$, $p = 2.4e\text{-}06$). The strength of hydrogen bonding in DESs is positively correlated with their viscosity. Our findings show that ChCl/OA and ChCl/CA exhibit higher viscosities than other DESs, implying a greater number of hydrogen bonds, which may enhance interactions with PMFs (Table 1). Furthermore, the π* parameter of DES exhibits a significant negative correlation with CE ($r = -0.64$, $p = 0.0043$). This finding indicates that merely increasing solvent polarity does not promote demethylation within this reaction system. We infer that this phenomenon is directly related to the hydrophobic nature of PMFs molecules. The PMFs molecules, possessing multiple methoxy groups (-$OCH_3$) in their backbone, exhibit overall lipophilicity (Duan et al., 2017). According to the principle of 'like dissolves like,' when the polarity of DES is excessively high, a polarity mismatch arises with the more hydrophobic PMF molecules. Consequently, we propose that DES systems with excessively high π* values actually diminish the solubility and dispersibility of PMF substrates. This poor solvation limits effective contact

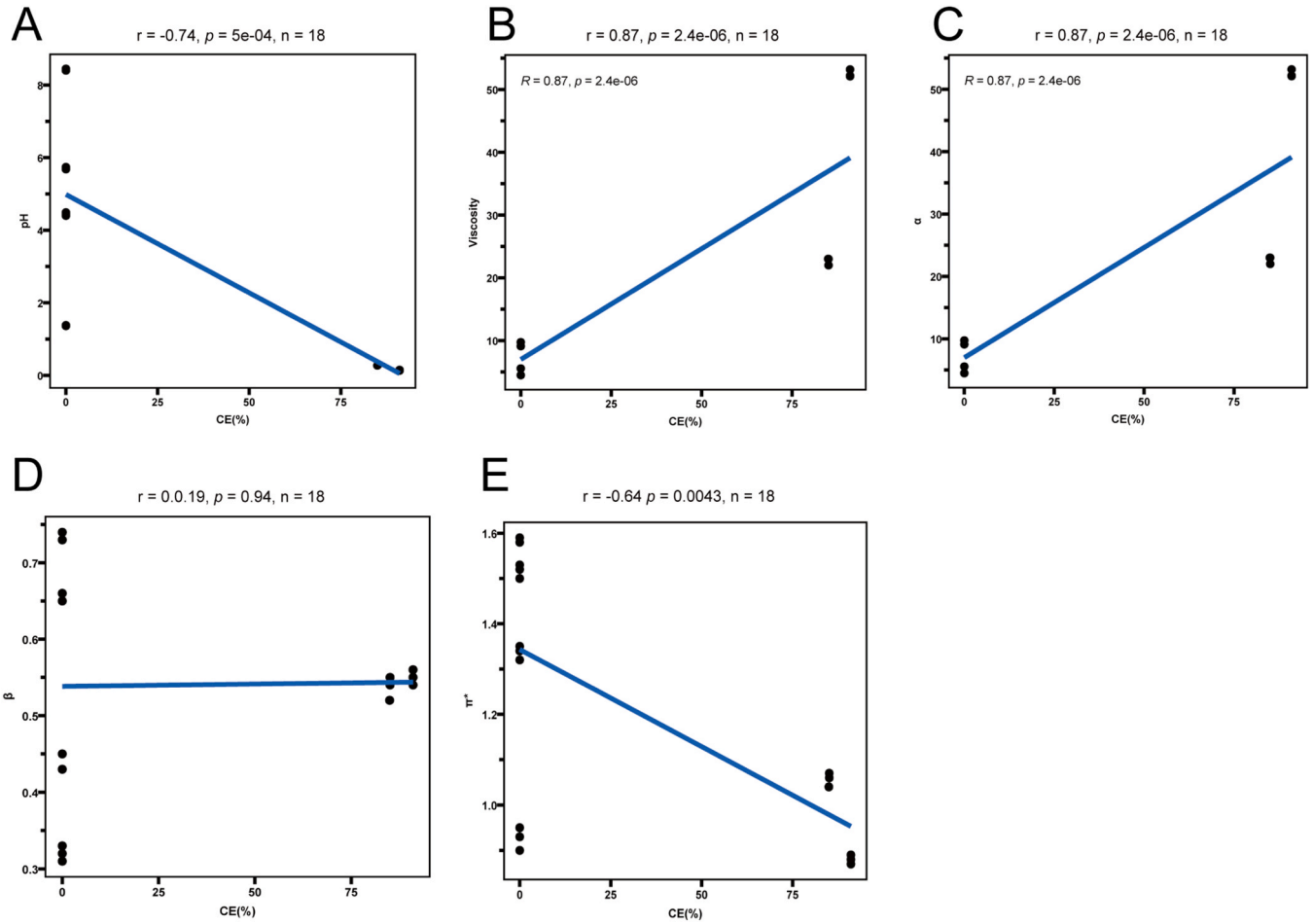



**Fig. 4.** Pearson correlation analysis of conversion efficiency with pH(A)、Viscosity(B)、 Kamlet-Taft-α(C)、Kamlet-Taft-β(D) and Kamlet-Taft-π*(E) of DES.

between substrate molecules and the catalytic active site within the DES, thereby inhibiting mass transfer of reactant molecules and ultimately reducing the efficiency of the demethylation reaction. The Kamlet-Taft α parameter exhibited an extremely high positive correlation with CE ($r = 0.87$, $p = 2.4e\text{-}06$), underscoring the importance of hydrogen bond acidity in enhancing demethylation. Higher α values indicate a stronger hydrogen bond donor capacity, enabling DESs to interact effectively with PMF methyl groups, facilitating methyl-oxygen bond cleavage, consistent with observations in 3.2.1. Conversely, the Kamlet-Taft β parameter showed a moderate positive but statistically insignificant correlation with CE ($r = 0.019$, $p = 0.94$), suggesting that hydrogen bond basicity has a relatively minor influence on demethylation. We further employed a statistical method independent of assumptions regarding data distribution—Spearman's rank correlation analysis. The results of the Spearman analysis demonstrated a high degree of consistency with those of the Pearson analysis, thereby reaffirming the correlation trends observed in our study (Supplementary Figure S3). Overall, these results highlight that pH, viscosity, Kamlet-Taft α, and π* parameters are critical factors that synergistically determine the efficiency of PMF conversion to 5-OH-PMFs within DES systems.

### 3.3. Optimization of Conversion Process Parameters

#### 3.3.1. Effect of Water Content in DES

Although DES1 demonstrated the highest overall conversion efficiency, safety concerns associated with oxalic acid necessitate focusing on citric acid-based DES2 in future studies to further optimize reaction conditions and enhance conversion efficiency. Citric acid-based DES are characterized by high viscosity, and the viscosity of the DES system can be reduced by adding water. However, water may significantly disrupt the intrinsic hydrogen bond network of the DES system, thereby reducing its conversion performance (Dai et al., 2015).Supplementary Figure S4 demonstrates that the addition of water significantly alters the hydrogen bonding within the original DES system. Within the 5–15 % water content range, the O–H band exhibits a shift towards lower wavenumbers, indicating that water molecules participate in the rearrangement of hydrogen bonds within the original system, revealing robust hydrogen bond formation. Upon further increasing the water content to 20 %, a distinct peak emerges at 3369 $cm^{-1}$ for the O-H band, indicating that the hydrogen-bond network has been reorganised and loosened (Supplementary Figure S4A). NMR data corroborates this phenomenon, confirming that the addition of water disrupts and reconfigures the hydrogen-bond network within DES2 (Supplementary Figure S4B). Thus, investigating the impact of water content on the conversion efficiency of DES is of critical importance. As shown in Fig. 5A, the conversion efficiency decreases significantly with increasing water content. When no water is added, the conversion efficiency reaches a peak (85.49 ± 0.07 %). However, when the water content reaches 5 % or higher, DES2 conversion efficiency significantly drops to near zero. As mentioned in the previous section, the conversion rate of PMFs is related to the hydrogen bond strength of DES. While water reduces the viscosity of the DES system, it also weakens the hydrogen bond strength of the DES, thereby reducing the conversion rate of PMFs. These findings highlight the critical role of water content in determining the conversion performance of DES2. Therefore, no additional water will be added to the DES system in subsequent optimization experiments.

#### 3.3.2. Effect of Material Ratio

Fig. 5B illustrates the effect of the PMFs/DES ratio on conversion efficiency. As the ratio increases from 1:5–5:1, the conversion efficiency

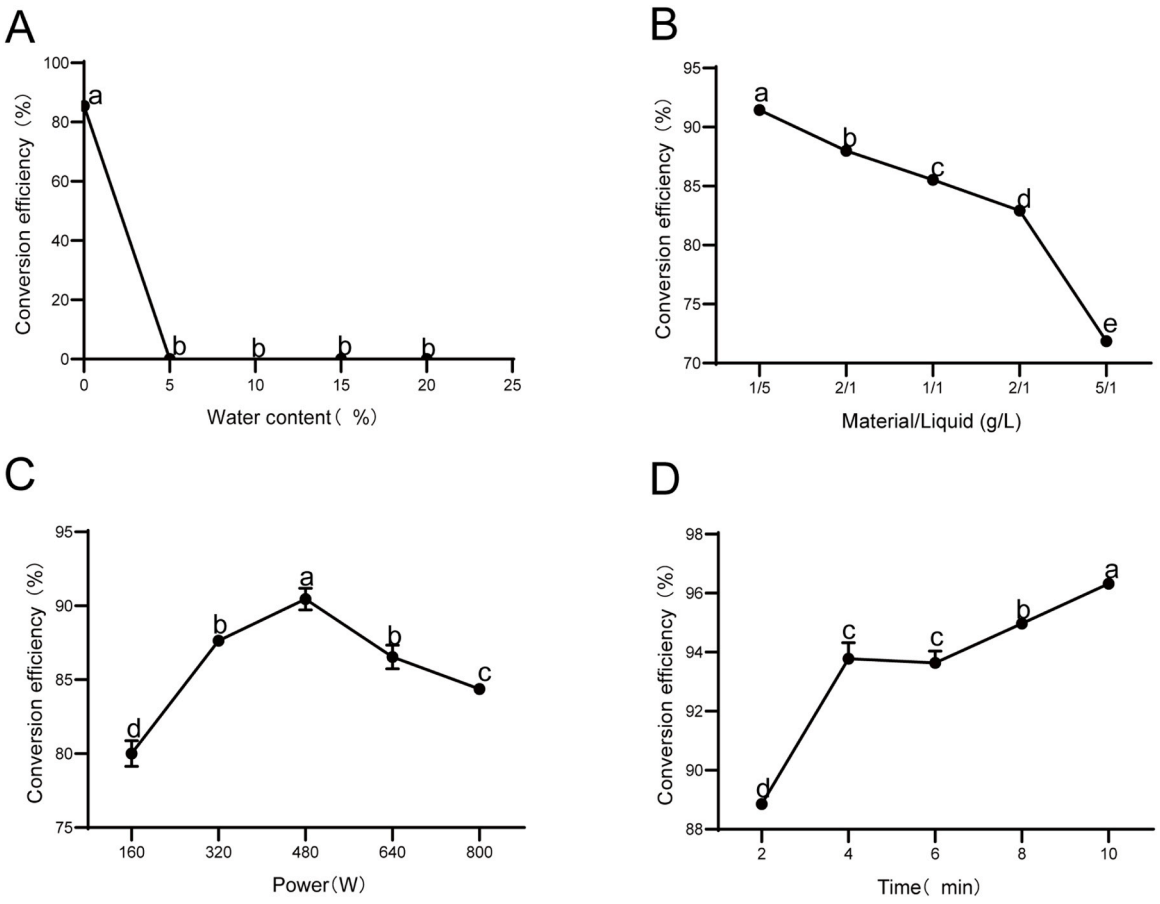



**Fig. 5.** Effect of DES water content(A), Material/liquid ratio(B), Power(C) and Reaction time(D) on the conversion efficiency of PMFs.

gradually decreases. Within the range of 1:5–1:1, the conversion efficiency decreases from 91.45 ± 0.07 % to 85.53 ± 0.02 %. However, when the ratio is further increased to 5:1, the conversion efficiency drops significantly to 71.87 ± 0.10 %. The solid-to-liquid ratio is a critical factor influencing the mass transfer characteristics of the reaction system. At lower solid-to-liquid ratios, the contact efficiency between substrate molecules and the solvent is higher, thereby promoting the reaction. Conversely, as the solid-to-liquid ratio increases, the available solvent per unit of solid decreases, limiting the sufficient interaction between the solvent and solute (Da Porto and Natolino, 2018). Additionally, increased solid content elevates the viscosity of the mixture, hindering solute mobility and impeding mass transfer processes (Zhang et al., 2019a, 2019b). Based on these findings, a lower solid-liquid ratio of 1:5 was selected as the optimized condition for subsequent experiments, as it provides a good balance between sufficient substrate-solvent interaction and controllable mixture viscosity, thereby ensuring optimal conversion efficiency.

#### 3.3.3. Effect of Microwave Power

As shown in Fig. 5C, the conversion efficiency follows a trend of initially increasing and then decreasing with changes in microwave power. At 160 W, the conversion efficiency reaches its minimum value (80.01 ± 0.88 %). Increasing the power to 480 W results in the highest conversion efficiency (90.47 ± 0.74 %, $p < 0.05$). However, further increasing the power to 800 W results in a significant decrease in conversion efficiency, dropping to 84.37 % ($p < 0.05$). This decline is likely attributable to the degradation of PMFs under excessive microwave power. High microwave power can generate intense localized 'hot spots' within the reaction system, leading to rapid and potentially uncontrolled temperature increases. Flavonoids, including PMFs, are known to be thermally sensitive (Huang et al., 2019). Therefore, while a certain level of microwave power is necessary to overcome the activation energy for the demethylation reaction, excessive power appears to be detrimental, causing compound degradation that outweighs the increase in conversion rate. Based on these findings, this experiment determined 480 W as the optimal microwave power, as it balances maximizing conversion efficiency with minimizing the risk of molecular degradation.

#### 3.3.4. Effect of Processing Time

Reaction time is a key parameter affecting conversion efficiency. The results indicate that extending the reaction time significantly affects conversion efficiency (Fig. 5D). At a reaction time of 2 min, the conversion efficiency was the lowest (88.85 ± 0.05 %). As reaction time increases, conversion efficiency gradually improves and reaches its maximum value at 10 min (96.32 ± 0.06 %, $p < 0.05$). Considering both conversion efficiency and cost-effectiveness, 10 min was determined to be the optimal reaction time.

Based on this, the optimal process parameters for the DES2 system to efficiently catalyze the hydroxylation of PMFs were determined as follows: a solid-to-liquid ratio of 1:5 (g/L), microwave power of 480 W, and a reaction time of 10 min. Under these optimized conditions, the conversion efficiency reached a maximum value of 96.32 ± 0.06 %. To comprehensively investigate the products, an untargeted metabolomics analysis was performed. The results confirmed that 5-DN and 5-DT were the most predominant 5-OH-PMFs. While trace amounts of other potential molecular were detected, their relative abundance was significantly lower than that of 5-DN and 5-DT (Supplementary Table S9). Therefore, subsequent quantitative analysis and mechanistic studies were focused on these two major compounds.

### 3.4. Evaluation of DES recycling performance

To evaluate the recycling capability of ChCl/CA, a series of solvent recovery and reuse experiments were performed under optimal conditions. The changes in conversion efficiency across recycling rounds are presented in Fig. 6. The results demonstrate that although the

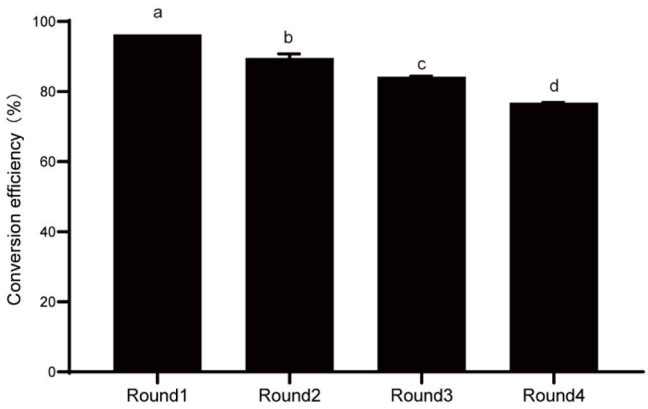



**Fig. 6.** The effect of DES recycling times on PMFs conversion efficiency.

conversion efficiency gradually declined during the recycling process, the DES retained substantial transformative activity over four cycles.

In the first round of use (Round 1), ChCl/CA exhibited excellent transformative performance, achieving a conversion efficiency of 96.32 ± 0.06 %, which was significantly higher than that observed in subsequent rounds ($p < 0.05$). After one cycle of recovery and reuse (Round 2), the conversion efficiency decreased to 89.59 ± 1.18 %. Further reuse resulted in conversion efficiencies of 84.25 ± 0.10 % in the third round (Round 3) and 76.81 ± 0.09 % in the fourth round (Round 4).

Notably, even after four cycles, the solvent retained over 75 % of its transformative efficiency, highlighting its reproducibility and potential value for industrial applications in the conversion of PMFs to 5-OH-PMFs. These results underline the practicality of the ChCl/CA DES system for sustainable and cost-effective processes in natural product modification.

### 3.5. Comparison with other methods

To date, various methods have been developed to hydroxylate PMFs into 5-OH-PMFs (Table 2). This study presents a novel microwave-assisted conversion system based on a deep eutectic solvent composed of choline chloride and citric acid (ChCl/CA). By optimizing process parameters, this method achieves significantly enhanced conversion efficiency. Compared with existing methods, this approach offers notable advantages, including environmental sustainability, high efficiency, and operational simplicity.

Traditional conversion methods employing boron tribromide ($BBr_3$) achieve high conversion efficiency but require toxic reagents and must be conducted under stringent conditions, such as low temperatures and the use of high-purity solvents. Additionally, these methods generate substantial quantities of hazardous byproducts during post-reaction processing, posing environmental and safety concerns (Li et al., 2007). The conversion method based on heating with hydrochloric acid, while requiring less specialized equipment, involves a prolonged reaction time of up to 24 h. Moreover, the corrosive and volatile nature of hydrochloric acid creates operational safety risks, which limit its feasibility for industrial applications (XU et al., 2015). In recent years, edible acid-phosphoric acid systems have gained attention due to their higher safety profile. However, such systems require a reaction time of 2 h and achieve a conversion rate of approximately 93 %, leaving room for improvement in both efficiency and product purification (Huang et al., 2019). The ChCl/CA DES system proposed in this study addresses many of the limitations associated with traditional methods. It eliminates the need for toxic reagents, significantly reduces reaction time, and operates under milder and safer conditions. Despite the absence of detailed economic cost-benefit analysis, the observed advantages in terms of higher yield, reduced processing time, and the use of greener, recyclable solvents suggest that the DES-based method holds significant potential

**Table 2**
Brief comparison of current and previous PMF demethylation methods.

| Solvent | Time | Condition | CE(%) | Reference |
|---|---|---|---|---|
| ChCl/CA | 10 min | microwave-assisted | 96.32 % (for both NOB and TAN conversion) | |
| $BBr_3$ | 30 min | −78 ℃ | 54 % for TAN | (Li et al., 2007) |
| HCl-EtOH(20:80,V/V) | 24 h | heated reflux | 99.5 % for TAN | (XU et al., 2015) |
| Phosphoric acid | 2 h | 175 ℃ | 93.71 % for TAN | (Huang et al., 2019) |

for economic viability in future industrial applications. A thorough techno-economic assessment at the pilot scale would be a valuable next step.

### 3.6. Bioactivity evaluation of 5-OH-PMFs

#### 3.6.1. Antioxidant Activity of 5-OH-PMFs

The chemical antioxidant capacity of 5-OH-PMFs were evaluated using ABTS, DPPH, and FRAP assays (Fig. 7A). The results revealed that 5-OH-PMFs exhibited significantly enhanced antioxidant activity compared to PMFs across all three assays. Specifically, the ABTS value for 5-OH-PMFs was 363.56 ± 3.85 μmol TE/g, significantly higher than the 50.22 ± 3.85 μmol TE/g observed for PMFs. Similarly, the DPPH value for 5-OH-PMFs was 239.49 ± 0.47 μmol TE/g, compared to 133.84 ± 1.23 μmol TE/g for PMFs, and the FRAP value for 5-OH-PMFs was 94.69 ± 0.51 μmol TE/g, markedly higher than the 30.38 ± 0.51 μmol TE/g for PMFs ($p < 0.001$ for all comparisons).

To further confirm the antioxidant activity of 5-OH-PMFs, an oxidative stress model was established in RAW264.7 cells using $H_2O_2$. Within the concentration range of 5–40 μg/mL, 5-OH-PMFs

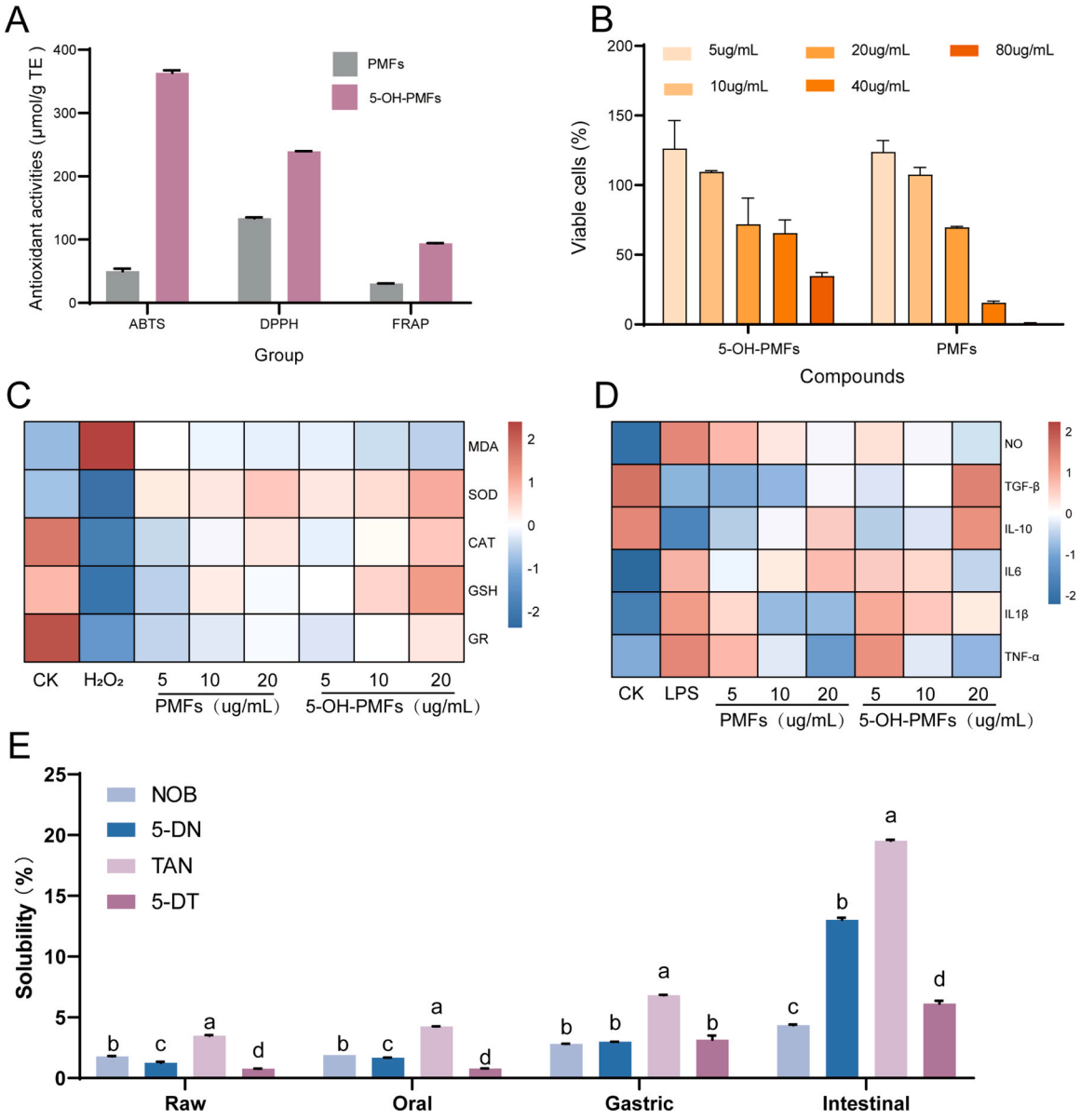



**Fig. 7.** Bioactivity evaluation of 5-OH-PMFs. A Chemical antioxidant activity. B Cytotoxicity. C Antioxidant activity in cell model. D Anti-inflammatory activity in cell model. E Solubility after simulated digestion in vitro.

demonstrated minimal cytotoxicity (Fig. 7B). At a concentration of 20 μg/mL, the antioxidant effect of 5-OH-PMFs was significantly stronger than that of PMFs. Specifically, 5-OH-PMFs significantly reduced MDA levels and enhanced the activity of SOD, CAT, GR, and GSH, bringing their levels comparable to those of the control group ($p < 0.05$) (Fig. 7C). The superior antioxidant activity of 5-OH-PMFs can be attributed to the structural modification of a hydroxyl group at the C5-position (Lai et al., 2015). The fundamental mechanism for radical scavenging by flavonoids involves the donation of a hydrogen atom from a hydroxyl group to neutralize a free radical (Lai et al., 2015; Wang et al., 2018). Specifically, the hydroxyl group at the C5-position plays a pivotal role. 5-OH group forms a strong intramolecular hydrogen bond with the adjacent carbonyl oxygen at the C4-position (Li et al., 2006). This intramolecular hydrogen bond not only stabilizes the parent molecule but, more importantly, stabilizes the phenoxyl radical formed after hydrogen donation. This stabilization effect lowers the bond dissociation energy of the 5-OH group, making it a more efficient hydrogen donor and thus enhancing the molecule's overall ability to scavenge free radicals and terminate oxidative chain reactions. These findings highlight the potential of 5-OH-PMFs as potent antioxidants, offering promising applications in functional foods or as therapeutic agents for oxidative stress-related diseases.

#### 3.6.2. Anti-inflammatory activity of 5-OH-PMFs

The anti-inflammatory activity of PMFs and 5-OH-PMFs were evaluated in a LPS-induced inflammatory model in RAW264.7 cells. As shown in Fig. 7D, both PMFs and 5-OH-PMFs inhibited nitric oxide (NO) production in a concentration-dependent manner. Notably, 5-OH-PMFs exhibited significantly higher efficacy, reducing NO levels were reduced to near baseline levelsat a concentration of 20 μg/mL. In terms of cytokine regulation, 5-OH-PMFs exhibited superior activity compared to PMFs. Specifically, 5-OH-PMFs significantly upregulated the expression of anti-inflammatory cytokines (such as TGF-β and IL-10) while effectively suppressing the expression of pro-inflammatory cytokines (such as IL-6, IL-1β, and TNF-α). These results highlight the potential of 5-OH-PMFs as a more potent anti-inflammatory agent than PMFs. Their ability to regulate inflammatory mediators and restore cytokine balance makes them a promising candidate for therapeutic strategies in the management of inflammatory diseases.

#### 3.6.3. In vitro digestion of 5-OH-PMFs

A three-stage in vitro digestion model (oral-gastric-intestinal) was employed to investigate the solubility changes of 5-OH-PMFs and PMFs throughout different stages of digestion. As depicted in Fig. 7E, the solubility of all tested compounds increased to varying extents following simulated gastrointestinal digestion. Among the three stages, intestinal digestion exerted the most pronounced solubilization effect, followed by gastric digestion, while oral digestion had a comparatively minor impact. Specifically, the solubility of 5-DN prior to digestion was 1.27 ± 0.09 %, which increased to 1.67 ± 0.02 % after oral digestion, further rose to 2.99 ± 0.007 % after gastric digestion, and significantly surged to 13.04 ± 0.16 % following intestinal digestion. Conversely, the solubility of NOB during the respective digestion stages was 1.78 ± 0.04 %, 1.89 ± 0.005 %, 2.82 ± 0.01 %, and 4.35 ± 0.05 %. Remarkably, the solubility of 5-DN post-intestinal digestion was approximately three times greater than that of NOB ($p < 0.05$). Interestingly, the solubility of TAN was significantly higher than that of 5-DT across all stages of digestion ($p < 0.05$), which contrasted the trend observed in the 5-DN/NOB comparison. These findings indicate that the influence of hydroxy substitution on solubility behavior is compound-specific.

The findings demonstrate that the intestinal environment exerts the most pronounced promotive effect on the solubility of flavonoids, aligning with the established understanding that flavonoids are predominantly absorbed in the intestine (Karn et al., 2021). The substantial increase in solubility observed during the intestinal digestion phase can likely be attributed to the solubilizing action of bile salts. These salts exhibit surfactant properties that facilitate the formation of micellar structures with hydrophobic flavonoids, thereby markedly enhancing their solubility in aqueous systems(Obeid et al., 2025).Furthermore, the distinct solubility profiles of 5-OH-PMFs and PMFs throughout digestion underscore the critical role of molecular structure in determining bioavailability. Typically, the hydroxyl group serves as a hydrogen bond donor, which strengthens interactions with polar solvents, thus potentially increasing solubility (Lee et al., 2023). However, 5-DT exhibits lower solubility compared to TAN, which may be attributed to its higher log P value (Supplementary Table S10). This increased lipophilicity is likely a result of hydrogen bonding interactions between the 5-hydroxy group and the adjacent carbonyl atom (Zhang et al., 2021). Elevated lipophilicity generally enhances a compound's intestinal membrane permeability, potentially contributing to increased biological activity in vivo.

## 4. Conclusion

This study demonstrated that employing strong carboxylic acid-based DES for the demethylation modification of citrus PMFs represents a green and sustainable strategy for enhancing the biological activity of PMFs. The pH, hydrogen bond strength, Kamlet-Taft α, and π* parameters of DES were identified as the key factors influencing PMF demethylation. Based on this, a green method was developed using a microwave-assisted choline chloride-citric acid (ChCl/CA) deep eutectic solvent system to catalyze citrus PMFs for the production of 5-OH-PMFs (primarily 5-DN and 5-DT). The optimal conversion efficiency of PMFs under this system reached 96.32 ± 0.06 %. Compared with traditional methods, this system significantly shortened the reaction time, demonstrating higher reaction efficiency and resource utilization. Additionally, 5-OH-PMFs exhibited markedly enhanced antioxidant and anti-inflammatory activities compared to their parent compounds, highlighting their potential applications in health-related products. However, the high viscosity of the DES system presents a challenge for operational simplicity. Fortunately, this limitation can potentially be mitigated through further optimization of the DES formulation. Additionally, the transformative mechanism of DES in the hydroxylation of PMFs warrants further investigation, which could provide valuable theoretical insights for designing customized DES formulations tailored to the preparation of natural bioactive compounds. In summary, this study provides a promising technical approach for the efficient preparation of 5-OH-PMFs and lays a solid foundation for their future development and application in bioactive product innovation.

## CRediT authorship contribution statement

**li shunjie:** Methodology, Data curation. **li yurong:** Writing – review & editing, Resources. **wei juanjuan:** Writing – review & editing, Resources, Methodology. **Zhiqin Zhou:** Writing – review & editing, Validation, Supervision, Funding acquisition. **ye zimao:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation.

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## Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

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## Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2025.122098.

## Data availability

Data will be made available on request.

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